K-Bound: When Is Label-Free Adaptation Knowable?

Supplementary Material

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This independently compiled supplement accompanies the current *K-Bound Short Main Paper*. It uses the same generated numerical authorities and records Appendices A–N: secondary evidence, proof dependencies, protocol ledgers, failure analyses, and reproducibility details. References to results stated in the main paper are descriptive here so this document does not depend on an external auxiliary file or on the main paper's page numbering.

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A Conditional Bridge from Cell Coverage to Population Coverage

The interval proposition covers its declared target. The following conditional argument explains what would be needed to change a cell-level target into a population one.

Additional condition for population transfer. For cell outcomes using the same loss as the population risk, such as accuracy and 0/1 loss, suppose that, on the same probability space,

$$\mathbb{P}(|\widehat{\Delta} - \Delta^{\text{cell}}| \le \varepsilon) \ge 1 - \alpha_{\text{cell}},$$
$$\mathbb{P}(|\Delta^{\text{cell}} - \Delta| \le b) \ge 1 - \delta,$$

where $b \ge 0$ is available without the new evaluation labels, then

$$\mathbb{P}(|\widehat{\Delta} - \Delta| \le \varepsilon + b) \ge 1 - \alpha_{\text{cell}} - \delta.$$

The triangle inequality applies on the intersection of the two coverage events, and the union bound controls its complement; independence is not needed. A population interval-supported action would use radius $\varepsilon + b$ and budget $\alpha_{\text{cell}} + \delta \le \alpha_{\text{target}}$. No such sampling-error bound is established for the reported panels, and their gates do not apply this enlarged radius. This conditional implication therefore adds no new population-safety claim to the empirical results.

For a nonlinear score such as macro-F1, an analogous argument must first define the population functional and a suitable sampling-error bound. Neither follows from the binary disagreement identity. Appendix L gives a counterexample with perfect cell-outcome coverage and negative population benefit.

B Additional Results and Non-Promoted Studies

This section preserves evidence outside the two primary protocols. Ordinary accuracy and balanced accuracy remain separate, and no result is pooled across datasets, candidates, or protocols. The input inventory is bound by source-manifest SHA-256 `03b1d2b1e9e5ed1cf835126f871ed83eb8497a1ac817` `27b3955af0d69eb89742`. Historical, incomplete, and invalid studies retain their original status.

B.1 Accuracy-Based Diagnostic Replays

Status: retrospective diagnostic.

Table 1 collects the remaining ordinary-accuracy replays; the dataset-specific interpretations follow below.

Table 1: Appendix ordinary-accuracy diagnostics. Each row is weighted over the evaluation unit named in the table; lower regret is better. Commitment counts ADAPT or FREEZE and is not interval coverage. The rows are retrospective or historical, not confirmatory. Office-Home's primary and stream-seed rows differ from its invalid five-checkpoint audit. ImageNet-C is candidate-specific. For PACS, $n = 12$ denotes equal-weight domain–seed regret units, whereas the archive contains 216 repeated action decisions.

Protocol	Status	n	KGA	Adapt	Freeze	$\mathrm{FA_u}$	Commitment rate
Office-Home primary	retrospective retention	35	0.0158	0.0468	0.0158	0.0000	0.3143
Office-Home stream seeds	descriptive replication	54	0.0215	0.0458	0.0217	0.0000	0.2778
ImageNet-C Tent	candidate-specific	135	0.0154	0.0191	0.0145	0.0222	0.0370
ImageNet-C EATA	candidate-specific	135	0.0024	0.0001	0.0342	0.0074	0.8074
ImageNet-C SAR	candidate-specific	135	0.0227	0.0529	0.0319	0.0000	0.2741
PACS (aggregate)	partial diagnostic	12	0.0431	0.0176	0.0446	0.0093	0.8148
CIFAR-10.1	locked diagnostic	48	0.0017	0.0190	0.0017	0.0000	0.4583

B.2 Balanced-Accuracy Diagnostic Replays

Status: retrospective diagnostic on a separate metric scale.

Table 2 reports the balanced-accuracy replays on their own metric scale, with each recorded candidate and evaluation unit kept distinct.

Table 2: Appendix balanced-accuracy diagnostics; lower regret is better. Each row uses the evaluation-cell weighting recorded by its protocol. The archived contracts name balanced accuracy for Camelyon17, ImageNet-R, and RxRx1; equality with ordinary accuracy can occur only under recorded class balance. ImageNet-R averages candidate backbones rather than one deployed gate. Camelyon17 OOD was opened, its B–v2 rows are separate within-seed diagnostics, and the displayed RxRx1 row is model seed 0. No interval or confirmatory claim is attached to this table.

Protocol	Status	n	KGA	Adapt	Freeze	$\mathrm{FA_u}$	Commitment rate
ImageNet-R backbones	architecture panel	480	0.0137	0.0064	0.0325	0.0125	0.4437
Camelyon17 OOD	opened one-sided	18	0.0000	0.0000	0.1381	0.0000	1.0000
Camelyon17 B–v2 Tent	opened diagnostic	108	0.0446	0.0097	0.0820	0.0000	0.2500
Camelyon17 B–v2 EATA	opened diagnostic	108	0.0260	0.0040	0.0911	0.0278	0.4722
Camelyon17 B–v2 SAR	opened diagnostic	108	0.0289	0.0016	0.1001	0.0093	0.4815
RxRx1 model seed 0	retention diagnostic	60	0.0000	0.2531	0.0000	0.0000	1.0000

B.3 Large-Scale Corruption: ImageNet-C

Status: retrospective candidate-specific diagnostic.

ImageNet-C is candidate-dependent. At the regenerated cross-fit authority, SAR has KGA/adapt/freeze regrets 0.0227/0.0529/0.0319 and 28/9/98 A/F/U decisions. It has a point advantage over both fixed policies and no false adaptation among 135 cells, but no seed-robust inferential support. Tent has regrets 0.0154/0.0191/0.0145 and 3/2/130 decisions; all three ADAPT decisions are false adaptations. EATA has regrets 0.0024/0.0001/0.0342 and 108/1/26 decisions, including one false adaptation. These rows support candidate-level diagnosis, not a dataset-wide success label. The main paper discloses the nonstandard SAR operating point and the absence of an official-setting control.

B.4 Office-Home

Status: retrospective diagnostic; the five-checkpoint route is invalid.

B.4.1 Valid Descriptive Records

The primary Office-Home row (Venkateswara et al., 2017) is conservative. KGA regret is 0.0158, tying always-freeze (0.0158) and below always-adapt (0.0468), with 0 adapt decisions. This is a descriptive point-estimate match that evaluates conservative retention rather than selective routing. The archived protocol does not supply a predeclared uniform stability guarantee for the full-fit versus out-of-fold benefit predictor.

The 54-cell Office-Home stream-seed replication in Table 1 has KGA regret 0.0215 versus 0.0458 and 0.0217, a small point advantage relative to each fixed policy. The three run seeds do not record independent checkpoint identities, and the seed-level interval includes zero, so the comparison remains descriptive.

Table 3: Historical Office-Home diagnostics with equal weight per recorded condition; lower accuracy regret is better. The rows are separate experiments. The stream-seed replay uses a fixed SAR candidate. The five-checkpoint route is invalid, so only its post-hoc fixed-SAR opportunity regret (.0027) and freeze regret (.0230) remain; its actions have no routing or interval-support meaning. A/F/U denotes ADAPT/FREEZE/ABSTAIN. No interval-supported claim is made.

Protocol	n	Router / Adapt / Freeze	A/F/U
Archived stream-seed replay	54	.0215 / .0458 / .0217	1/14/39
Five-checkpoint candidate audit	15	invalid / .0027 / .0230	n/a

B.4.2 Invalid Five-Checkpoint Route

The post-hoc five-checkpoint Office-Home study is a candidate opportunity audit, not a valid KGA result. It uses five independently trained source checkpoints, the three target-test domains, one fixed IID/tiny condition, and Tent, EATA, and SAR. Checkpoint identity passed, but routing and serialization did not. The identifying agreement relation is binary-only and does not apply to the 65-class task; the old spectral estimate was sign-indeterminate and unbounded. Thirteen of 15 anchors were negative, 9 of 60 stored $\widehat{b}$ values fell outside $[0, 1]$, and EATA and SAR produced exactly the same predictions in 10 of 15 cells. The historical leave-one-out audit has only three cells per checkpoint, whereas a 90% empirical order-statistic single-candidate radius requires at least 10 total cells. All five result files fail strict JSON and contain 57 literal `Infinity` values in total.

The archived exploratory implementation printed ABSTAIN in all 15 cells, but those outputs have no safety, routing, or interval-support meaning. The valid descriptive quantities are checkpoint accuracies and candidate opportunity: freeze is 0.5711, SAR is 0.5914, their paired difference is 0.0203 (nominal post-hoc 95% t-interval $[0.0123, 0.0283]$), and the per-cell oracle is 0.5941. Oracle minus SAR is 0.0027 (nominal post-hoc 95% interval $[-0.0024, 0.0077]$). The best adapted candidate improves over freeze in 13 cells, ties in one, and harms in one. Because the target was already opened and SAR was chosen after comparing the audit means, these statistics are post-hoc opportunity evidence only.

B.5 Camelyon17

Status: opened, one-sided retrospective diagnostic.

The archived Camelyon17 OOD evaluation (Bandi et al., 2019) gives zero KGA and always-adapt regret, versus 0.1381 for freeze. All 18 cells are helpful. This opened panel is therefore a one-sided descriptive diagnostic, not prospective evidence of selective routing. The B–v2 within-seed diagnostics are separate. Their regenerated Tent/EATA/SAR KGA regrets are 0.0446/0.0260/0.0289, versus always-adapt regrets 0.0097/0.0040/0.0016 and always-freeze regrets 0.0820/0.0911/0.1001. None has lower regret than both fixed policies. The three run seeds do not identify independently trained checkpoints, and the target was already open. These rows remain stress diagnostics, not untouched hospital-domain evidence.

B.6 RxRx1

Status: retrospective retention diagnostic.

The displayed RxRx1 row (Sypetkowski et al., 2023) is model seed 0: KGA and freeze regret are zero against 0.2531 for always-adapt, with no adapt decisions. A separate audit across three independently trained model checkpoints repeats the same all-freeze tie. It strengthens model-variation reproducibility but adds no adapt exposure. RxRx1 therefore evaluates reproducible conservative retention rather than selective routing.

B.7 iWildCam

Status: invalid current metric contract; no canonical numerical row.

iWildCam (Koh et al., 2021) is not assigned a canonical numerical row. The archived scorer used sklearn macro-F1, which averages prediction-only classes, rather than the official WILDS metric, which averages labels present in the target truth. The archived runtime, sample identities, and target population were not sealed under the current contract, so its diagnostic scores and routing actions remain audit-only. A pinned rerun with the official metric, sealed sample identifiers, and a population hash is required before promotion. The older binary dominance flag and the later replay under the same invalid metric contract remain hash-locked only as superseded audit history.

B.8 Boundary and Transfer Diagnostics

Status: retrospective transfer diagnostics.

B.8.1 PACS

PACS (Li et al., 2017) KGA regret is 0.0431, compared with 0.0176 for always-adapt. The regret summary has 12 equal-weight domain–seed units. Its 216 action records are repeated decisions within those units, not 216 independent regret observations. The archive lacks the residual detail needed to replay the current gate, so PACS remains an aggregate performance diagnostic.

B.8.2 ImageNet-R Backbone Breakdown

ImageNet-R (Hendrycks et al., 2021) KGA regret is 0.0137, compared with 0.0064 for always-adapt and 0.0325 for always-freeze, across the ten-backbone architecture panel. No backbone has lower KGA point regret than each fixed comparator. KGA regret is higher than always-adapt regret on 8 of 10 backbones; four of those eight have a 0% harmful base rate, so routing has no harm to avoid. On `efficientnet_b0` (100% harmful) and `swin_t` (60.4% harmful), KGA improves over always-adapt but exactly matches always-freeze and makes no adapt decisions. The pattern is consistent with the regime map: routing can add value only where candidate harm is present, while these two rows evaluate conservative retention rather than selective routing.

B.8.3 CIFAR-10.1

CIFAR-10.1 (Recht et al., 2018) matches always-freeze at 0.0017 and is retained as a low-margin cross-seed transfer diagnostic. This pattern alone does not prove structural non-identifiability; weak evidence, low margin, estimator inadequacy, or calibration transfer can produce the same behavior.

B.9 Historical Protocol-Matched POEM and AETTA Ports

Status: historical local-port audit; not a synchronized official comparison. The repository retains one historical head-to-head artifact on the CIFAR-10-C Tent stream. Its KGA, POEM-style, and AETTA-style regrets are 0.00157361, 0.00880463, and 0.00732986; the associated always-adapt and always-freeze regrets are 0.00792338 and 0.12409792. Historical KGA-minus-port paired differences and unadjusted 95% paired-bootstrap intervals are

Port	Difference	95% interval
POEM-style	−0.00723102	[−0.00878038, −0.00570806]
AETTA-style	−0.00575625	[−0.00714260, −0.00439859]

Holm adjustment applies only to the archived p-values; the adjusted value for each of these two comparisons is 0.00039996.

Those estimates are protocol-matched diagnostics from local ports, not results from untouched official implementations. They also use an earlier KGA policy and are not synchronized with the current three-way cross-fit. The numerical ordering is therefore historical audit evidence only: it does not establish current-policy or official superiority. The source artifact is `experiments/kbound/results/mixed_headtohead_v1/HEADTOHEAD_RESULTS_cifar10c_tent_primary.json`; its stored `WIN` and `kga_beats` labels are withdrawn from the current claim set. A defensible new comparison would replay all methods on identical current-policy records and freeze the multiplicity family before scoring.

B.10 So2Sat Development and Action-Unit Audit

Status: development stop before target access.

The locked So2Sat-LCZ42 v1 study screened Tent and SAR in a development-only, multi-city, multi-checkpoint candidate-feasibility stage. It was not a target evaluation. Tent showed both helpful and harmful development cases, but neither candidate passed every predeclared eligibility check. The locked status is therefore *no feasible candidate; stop before gate calibration.* The detailed development authority and its binding receipt remain outside the default public build closure; the public release records only the protocol shape, stopping decision, and access boundary.

No held-out gate-calibration rows or target inputs, pixels, or labels were accessed. The study therefore contains no target natural-shift score. The gate-fit outcomes are open, so retuning on them cannot create fresh confirmatory evidence. V1 also declares one target action per city while its target runner creates city–checkpoint actions. This mismatch did not affect the stop because no target action was created. Prospective v2 resolves the action unit as city–checkpoint and supplies a source-complete runner. It remains a no-go: no v2 pre-calibration seal or target authorization exists, and gate calibration and target evaluation remain unopened. V2 does not retroactively change the negative v1 record.

B.11 Withdrawn Development Proxy for the Population Budget

Status: negative retrospective diagnostic; estimation route withdrawn.

An unlabeled deployment batch cannot verify $|\gamma| \leq \beta$, and $\beta = 0$ is not a safe default. The tested source-like development proxy was 1.4–50× too small on real evaluation cells: 24–73% fell outside $\mathcal{C}_\beta$, commit error was 0.4–16.6%, and five of ten configurations returned zero commitments on all 405 ImageNet-C cells. The proxy did not observe the theorem's disagreement-conditional residual, and its named raw artifact is not regenerable from the canonical bundle. We therefore withdraw estimation and require domain knowledge, a declared coupling, or historical labeled deployments under a transfer assumption.

The population layer targets Δ, whereas empirical calibration targets Δ^{cell}. Appendix A states the additional sampling argument needed to bridge them. Neither layer establishes risk alignment, estimator transfer, or residual coverage in a new domain.

C Compact Assumption and Claim Ledger

Table 4 separates formal claims from empirical findings and records the conditions and scope limits for each.

Table 4: Formal half of the claim-to-support ledger. Each row separates its target and assumptions from what is and is not supported.

Statement and target	Required assumption or protocol	Supported conclusion	Not supported
Disagreement reduction; population Δ	binary labels, 0/1 loss, fixed pair, $\mu_T(D) > 0$	$\Delta = 2\mu_T(D)(M + \gamma)$	calibration transfer or observability of γ
Interior ambiguity; population Δ	Main-paper Thm. 1; $\beta > 0$, $\|M\| < \beta$, fibre richness	evidence-identical laws with opposite nonzero benefit	ambiguity outside the declared class
Closed band; population Δ	Main-paper Prop. 1; strict semantics and richness	no uniformly sound strict action on $\|M\| \leq \beta$	unequal task risk at every boundary law
Strict frontier; population Δ	Main-paper Thm. 2; fixed evidence and declared $\mathcal{C}_\beta$	pointwise maximal sound three-action rule	learning β from the unlabeled batch
Audit floor; residual radius	common observable-law fibre and uniform validity	$\widehat{\beta} \geq \Gamma_z$ with the stated probability	a finite-batch estimator or data-learned β
Coverage-to-action; declared cell target B	Main-paper Prop. 2 and marginal coverage	marginal false ADAPT and false FREEZE control for B	population-risk, conditional, group, or anytime control

Table 5: Empirical and future-evidence half of the claim-to-support ledger. Evaluation units and statuses remain protocol-specific; no row upgrades retrospective evidence.

Statement and target	Required assumption or protocol	Supported conclusion	Not supported
CIFAR-10-C; measured cells	opened one-checkpoint grid and outcome-disjoint cross-fit	candidate-specific actions, errors, and point regret; Tent nominal intervals favorable	prospective confirmation; six-contrast Holm rejection
CCT-20; measured cells	locked 45-cell protocol and declared utility endpoint	all-frozen utility preservation with 0/44/1 actions	selective routing, improvement over always-freeze, or population safety
Other natural replays; measured cells	Tables 1–2; fixed splits and candidates	protocol-specific retention or adverse behavior	one pooled success claim
So2Sat; development eligibility	Main-paper development stop; locked feasibility rules	no candidate passed the pre-target screen	gate calibration or a target score
ImageNet-C; measured cells	Table 1; candidates and operating points kept separate	candidate-specific diagnostic evidence	dataset-wide or published-default conclusion
Other replay scope; recorded units	Tables 1–2	observed protocol behavior	structural unknowability or universal transfer
Prospective confirmation; future targets	Sec. K; diverse unopened environments	a frozen evaluation design	a current empirical result
Physical camera; future sessions	locked session split and held-out recordings	protocol infrastructure only	a completed camera result

D Calibration and Evaluation Contracts

D.1 Controlled-Grid Contract

The CIFAR-10-C grid is an opened, dependent, constructed six-family diagnostic. Its fitting unit is the condition cell. Stable identifiers assign deterministic score folds. For each fold, its complement is divided into disjoint estimator-fit and residual-calibration subsets. The fitted predictor and exact-rank radius are then applied only to the score fold. Consequently, the scored outcome $B_i = \Delta_i^{\text{cell}}$ enters neither the fit nor the calibration set that determines its own prediction, radius, or action. This is a three-way cell-outcome-disjoint cross-fit, not leave-one-cell-out calibration.

The CIFAR-10-C grid uses one ResNet-18 checkpoint and five RNG stream/run seeds. Within each seed it crosses six corruption families, severities 1 and 5, batch sizes 200/16/8, IID/imbalanced/single-class composition, two update schedules (10 steps at 10^{-3} and 50 steps at 2.5×10^{-3}), and two repeats: 432 cells per seed and 2,160 per candidate. Point summaries weight cells equally. The family analysis first averages seeds and cells within a family and then weights the six family means equally. These are opened, dependent, constructed cells coupled through one corpus and checkpoint; the cross-fit does not make them exchangeable or validate coverage on unseen environments.

Tent, EATA, and SAR are local protocol-matched ports. Candidate preview runs in training mode with BatchNorm running statistics disabled, so the measured delta combines evaluation-batch statistic replacement with gradient updates to normalization affine parameters. The two recorded update schedules are part of the grid. They are not untouched reproductions of each method's published default operating point; the main text gives the additional SAR-specific departure.

D.2 Natural-Shift Contract

Development conditions fit the benefit model; a disjoint calibration partition supplies residuals; choices are frozen; and the target partition is scored once where the archived protocol records all three stages. When the saved evidence does not support that complete chain, the row is marked descriptive or diagnostic. Replaying a table does not strengthen its original design.

The candidate TTA step itself is transductive. Episodic candidates update on each unlabeled evaluation batch before predicting that batch. Online candidates update on a separate auxiliary stream, but prediction still uses evaluation-batch BatchNorm statistics. Thus "disjoint" above refers to development, residual-calibration, and target partitions (and to the recorded auxiliary stream versus evaluation sample identities); it does not mean that candidate inference is inductive. No target labels enter adaptation, prediction, evidence, or the action.

D.3 Action Contract

Every action is computed from Z, $\widehat{\Delta}$, and ε without target labels. Offline labels determine Δ^{cell}, the measured oracle action, empirical regret, and false-commit indicators only after the action is fixed. Existing artifact fields named `delta` store this cell outcome; their schema and numerical values are unchanged. Abstention resolves to the frozen prediction for measured task-performance accounting, while its log retains the distinction from an interval-supported FREEZE decision.

The whole-pipeline invariant is tested by perturbing one scored B_i and rerunning all fits. Its $\widehat{\Delta}_i$, ε_i, and action must remain unchanged. The maintained test `tests/test_controlled_grid_crossfit.py` checks this for the implementation and also exhibits the old radius leak as a negative control. This is a software non-leakage result for the scored cell; it does not prove exchangeability, estimator validity, or future-domain coverage.

D.4 Configuration Contract

A configuration identity binds the checkpoint, adapter and update settings, evidence schema, benefit estimator, residual construction, α, and action unit. Section H records that identity once. Any mismatch requires a declared transfer argument or recalibration; otherwise the controller does not commit.

E Decision Exposure and Repetition Audit

Point regret alone can conceal whether a conservative match exercised the ADAPT branch. Table 6 therefore reconstructs action counts from the same canonical panel used in the main paper's primary CIFAR table and Tables 1 and 2. A/F/U denote adapt, freeze, and abstain. PACS is omitted from the action table because the released aggregate cross-checks regret but does not contain the per-cell benefit estimates and residuals required for a complete gate replay. We omit iWildCam because its archived actions were scored under the wrong macro-F1 contract.

Table 6: Action exposure in replayable rows. A/F/U denotes ADAPT/FREEZE/ABSTAIN. Counts are unweighted evaluation cells, not independent experiments; regret tables use the weighting declared by each protocol. False A/A and false F/F use the corresponding action denominator. A dash means that the direction was unexposed or was not reconstructed by the maintained authority; it does not mean zero error. Controlled-grid rows use the current three-way cross-fit; the remaining rows retain their stated diagnostic protocol. The canonical numerical panel is bound to SHA-256 `35d4c165843de1ece3cb35ffb4ac50dbfcbfa33b646754e2b6d133fbdfa78c6e`; its input inventory is bound separately by the source manifest.

Row	n	A	F	U	false A/A	false F/F	repetition / scope
Office–Home primary	35	0	11	24	–	–	2 test-stream seeds; ties freeze
Office–Home stream-seed repl.	54	1	14	39	0/1	–	3 test-stream seeds; point-only edge
ImageNet–C SAR	135	28	9	98	0/28	–	5 seeds × 27 cells
ImageNet–C Tent	135	3	2	130	3/3	–	5 seeds × 27 cells
ImageNet–C EATA	135	108	1	26	1/108	–	5 seeds × 27 cells
ImageNet–R	480	179	34	267	6/179	–	4 seeds × 10 backbones × 12 cells
CIFAR–10–C Tent	2160	1091	337	732	2/1091	1/337	5 seeds × 432 cells
CIFAR–10–C EATA	2160	1207	105	848	0/1207	0/105	5 seeds × 432 cells
CIFAR–10–C SAR	2160	1414	0	746	0/1414	–	no FREEZE exposure
CCT–20	45	0	44	1	–	1/44	5 checkpoints × 9 locations
Camelyon17 OOD	18	18	0	0	0/18	–	opened helpful-only OOD subset
Camelyon17 B–v2 Tent	108	27	0	81	0/27	–	3 within-seed grids; diagnostic
Camelyon17 B–v2 EATA	108	51	0	57	3/51	–	3 within-seed grids; diagnostic
Camelyon17 B–v2 SAR	108	52	0	56	1/52	–	3 within-seed grids; diagnostic
RxRx1 J	60	0	60	0	–	–	model seed 0; 3-checkpoint audit also all-freeze
CIFAR–10.1 K	48	0	22	26	–	–	locked cross-seed replay

The largest adapt exposure occurs on CIFAR-10-C, making it the main diagnostic of the adapt branch. Tent records two false adaptations among 1,091 ADAPT decisions and one false FREEZE among 337 FREEZE decisions; EATA and SAR record no false adaptations, although SAR never exposes FREEZE. The CCT-20 row supplies the complementary all-frozen case: one false FREEZE among 44 FREEZE decisions and no ADAPT exposure. Primary Office-Home and RxRx1 have zero adapt exposure, while the 54-cell Office-Home replication has one adapt decision. Those rows chiefly test conservative retention. The invalid five-checkpoint Office-Home route and the invalid-metric iWildCam archive are excluded. Camelyon17 OOD has the opposite limitation: every archived cell is helpful and every action adapts. On ImageNet-C, Tent makes three ADAPT decisions and all are false adaptations, whereas EATA makes 108 ADAPT decisions with one false adaptation and SAR makes 28 with none. Similar regret values can therefore hide different operating behavior.

The repetition column also prevents a common inflation of evidence. Five seeds of one checkpoint and grid protocol are not five independently designed studies. Ten ImageNet-R backbones are a panel of candidate policies, not ten target domains for one frozen gate. Three Camelyon17 within-seed grids permit a stability diagnostic but do not create an untouched hospital-domain replication. RxRx1 is stronger on model variation because the source checkpoint is independently trained three times, but its all-freeze behavior still provides no adapt exposure.

F Inference Discipline

F.1 Point Estimates, Paired Gaps, and Clusters

For a baseline $b \in \{\text{always-adapt}, \text{always-freeze}\}$, the paired routing gap at condition j is

$$G_{bj} = \mathrm{regret}_{b,j} - \mathrm{regret}_{\mathrm{KGA},j}\,.$$

A positive mean gap means lower regret for KGA. All paired routing-gap intervals reported in this paper use this baseline-minus-KGA sign convention. The point-estimate two-comparator criterion requires both means to be positive. Interval support additionally requires the lower confidence limit for both paired means to exceed zero at the declared resampling unit. These definitions are intentionally separate: Table 1 can establish a point ordering, but not an inference claim without the associated unit, interval artifact, and justified coverage assumptions. A positive computed lower endpoint is numerical interval support under the resampling model, not verified coverage. Multiplicity adjustments do not repair invalid marginal intervals or tests.

Condition-level resampling treats cells as exchangeable and can understate uncertainty when severities and compositions share a corruption family. Corruption-family resampling preserves observed within-family dependence, but few clusters limit reliability; conservativeness is not guaranteed. Seed resampling measures variation across repeated training or test streams only when the seeds represent the intended deployment randomness. The compact paper therefore does not use one generic "bootstrap CI" label. An inference claim must state whether the unit is cell, corruption–severity pair, corruption family, seed, backbone, domain, or physical session.

For a vector of cluster gaps $G = (G_1, \ldots, G_m)$, the exact sign-flip reference calculation uses the joint null condition

$$G \overset{d}{=} s \odot G \quad \text{for every } s \in \{-1, +1\}^m. \tag{1}$$

Independent zero-symmetric cluster gaps are sufficient. Exchangeability, marginal symmetry, or one global sign symmetry alone is not sufficient. Enumeration removes Monte Carlo error but does not establish this null condition.

F.2 Current-Policy Corruption-Family Sensitivity

The retrospective family contains all six candidate-by-baseline contrasts. The replay policy, corruption-family unit, sign-flip reference test, and Holm adjustment were selected after the underlying CIFAR outcomes were open. The current-policy artifact averages run seeds and condition cells within each of the six families. Tent's nominal intervals favor KGA against always-adapt ($[0.002087, 0.009537]$) and always-freeze ($[0.062822, 0.177573]$). EATA's adapt-side interval crosses zero ($[-0.001348, 0.004382]$), although its freeze-side interval is positive. SAR is worse than always-adapt. These nominal bootstrap intervals are distinct from the retrospective sign-flip/Holm analysis. The within-Tent two-contrast Holm calculation is itself post hoc and is not the six-test family reported in Table 7.

Table 7: Retrospective sensitivity over six opened CIFAR-10-C corruption families. Each candidate uses 2,160 cells from one checkpoint; seeds and conditions are averaged within family, and the six family means receive equal weight. Entries are baseline regret minus KGA regret with nominal, unadjusted 95% family-bootstrap intervals; positive values favor KGA. Holm values adjust the six retrospective sign-flip reference tests. For Tent, the post-hoc within-candidate Holm values are 0.046875 (adapt) and 0.03125 (freeze), while the six-contrast values are 0.140625 and 0.09375. The replay, unit, tests, and adjustment were not preregistered. The table supplies no verified coverage, independent-checkpoint inference, prospective confirmation, natural-shift evidence, or official POEM/AETTA comparison.

Candidate	Adapt gap [95% CI]	Freeze gap [95% CI]	Six-way Holm p (A/F)
Tent	0.00606 [0.00209, 0.00954]	0.12202 [0.06282, 0.17757]	0.14062/0.09375
EATA	0.00164 [-0.00135, 0.00438]	0.12970 [0.06893, 0.18746]	0.31250/0.09375
SAR	-0.00145 [-0.00294, -0.00027]	0.13869 [0.07972, 0.19683]	0.90625/0.09375

F.3 Retrospective Interval Inclusion by Family

This audit summarizes the stored per-cell authority produced by the three-way cross-fit. For every scored cell, its measured outcome was absent from both estimator fitting and residual calibration. The audit performs no tuning, new target access, or training; it verifies the stored predictions, radii, actions, and input/code/output digests.

The observed inclusion counts are 1,976/2,160 for Tent, 1,950/2,160 for EATA, and 1,949/2,160 for SAR. Tent also has one false FREEZE among 337 FREEZE decisions. These are in-sample cross-fit diagnostics on opened, dependent cells, not independent estimates for new environments. Family rows localize misses and widths but do not supply group-conditional guarantees. The nominal 90% setting is therefore not a validated coverage promise.

Table 8: Overall retrospective interval diagnostics for 2,160 equally weighted cells per CIFAR-10-C candidate. Inclusion concerns measured cell benefit, not population benefit. Marginal directional errors divide by all cells; conditional errors divide by the corresponding action count and are undefined when that action is absent.

Candidate	n	Nominal	Inclusion	Mean width	Median width	Commitment	$\mathrm{FA}_u/\mathrm{FA}_c$	$\mathrm{FF}_u/\mathrm{FF}_c$
TENT	2160	0.9000	0.9148	0.0478	0.0457	0.6611	0.0009/0.0018	0.0005/0.0030
EATA	2160	0.9000	0.9028	0.0397	0.0389	0.6074	0.0000/0.0000	0.0000/0.0000
SAR	2160	0.9000	0.9023	0.0286	0.0290	0.6546	0.0000/0.0000	0.0000/–

Table 9: Current-policy interval and decision diagnostics by corruption family. Each row aggregates equal-weight opened cells after the three-way cross-fit; the scored cell outcome enters neither its fit nor calibration set. Width is 2ε, and A/F/U means ADAPT/FREEZE/ABSTAIN. Conditional errors use the corresponding action count as denominator and are undefined when that action is absent. The table is retrospective and is not held-out group-coverage validation.

Candidate	Family	n	Inclusion	Mean width	Commitment	False A/A; FA_c	False F/F; FF_c
TENT	contrast	360	0.8361	0.0478	0.6917	0/180; 0.0000	0/69; 0.0000
TENT	defocus_blur	360	0.9417	0.0480	0.7083	0/180; 0.0000	0/75; 0.0000
TENT	fog	360	0.9472	0.0480	0.6917	0/180; 0.0000	0/69; 0.0000
TENT	gaussian_noise	360	0.8667	0.0480	0.8556	0/305; 0.0000	1/3; 0.3333
TENT	jpeg_compression	360	0.9694	0.0478	0.3472	2/66; 0.0303	0/59; 0.0000
TENT	pixelate	360	0.9278	0.0475	0.6722	0/180; 0.0000	0/62; 0.0000
EATA	contrast	360	0.8417	0.0396	0.5694	0/180; 0.0000	0/25; 0.0000
EATA	defocus_blur	360	0.9278	0.0398	0.5694	0/180; 0.0000	0/25; 0.0000
EATA	fog	360	0.9139	0.0399	0.5750	0/180; 0.0000	0/27; 0.0000
EATA	gaussian_noise	360	0.8528	0.0399	0.9722	0/350; 0.0000	0/0; –
EATA	jpeg_compression	360	0.9694	0.0396	0.4056	0/137; 0.0000	0/9; 0.0000
EATA	pixelate	360	0.9111	0.0393	0.5528	0/180; 0.0000	0/19; 0.0000
SAR	contrast	360	0.8694	0.0287	0.5000	0/180; 0.0000	0/0; –
SAR	defocus_blur	360	0.9306	0.0287	0.5000	0/180; 0.0000	0/0; –
SAR	fog	360	0.8944	0.0283	0.5000	0/180; 0.0000	0/0; –
SAR	gaussian_noise	360	0.8417	0.0286	1.0000	0/360; 0.0000	0/0; –
SAR	jpeg_compression	360	0.9667	0.0283	0.8944	0/322; 0.0000	0/0; –
SAR	pixelate	360	0.9111	0.0288	0.5333	0/192; 0.0000	0/0; –

F.4 False-Adapt and False-Freeze Accounting

The event in the main paper's false-adapt definition is unconditional and concerns measured cell benefit. Its reported frequency divides ADAPT decisions with $\Delta_j^{\text{cell}} \leq 0$ by all evaluated units, including freezes and abstentions. This need not estimate the population-harm event $\{g = \textsc{adapt}, \Delta \leq 0\}$. Conditional false-adapt divides by adapt decisions and is undefined when no adaptation occurs. A zero observed count should be accompanied by the number of adapt decisions; otherwise a controller that never adapts appears perfect. Table 6 exposes that denominator.

False FREEZE is the symmetric event: the policy chooses FREEZE when measured cell benefit is nonnegative. Its marginal rate divides by all evaluated units, whereas its conditional rate divides by FREEZE exposure. The latter is undefined when the policy never freezes, as in the current SAR grid. Every directional-error statement therefore appears with its action denominator.

The three-way split removes the direct path from B_i to its own prediction, radius, and action. It does not make the overlapping fitted models or the constructed cells independent. Pooled cross-fit inclusion therefore remains a descriptive reuse of the opened panel and supplies neither exchangeability nor a finite-sample guarantee for the next corruption family. A genuinely held-out family or environment remains a different protocol.

F.5 Multiplicity and Search History

The repository contains many exploratory configurations. One superseded census reported 1,387 determinations and 326 positives (23.5%); a broader archived scan reported 1,427 determinations and 345 positives (24.2%). Neither census froze one family definition, file list, and commit, so neither is the release multiplicity family. The reconciled panel is also post hoc, although some component protocols were individually locked. Confirmatory work must declare every candidate–track test, calibration and target unit, seed structure, inference unit, and two-comparator criterion before labels are opened, while retaining failures and incomplete runs.

G Track-Level Provenance and CCT Ledger

Separate archival records preserve two additional scope checks outside the canonical panel. FMoW Protocol L reached held-out scoring and matched always-freeze, while always-adapt had lower regret; its status remains not-cleared. PovertyMap stopped at its preregistered development screen because the harm-detection AUC was 0.637, below the 0.65 threshold, so held-out validation and test scoring did not run. These records remain in the claim inventory and are not pooled with the canonical rows.

The provenance table is part of the result, not administrative metadata. A reviewer can trace a canonical row to compact records and their original archive hashes. If the original large artifact is not shipped, the manifest still records its byte count and digest. This guards against silent replacement of a result while allowing the release to omit raw datasets and oversized transient checkpoints.

Table 10: Release-status inventory, not a performance comparison. "Replayable" means stored quantities suffice to recompute the displayed gate output under the named scorer; it does not imply fresh data, independent units, or confirmatory status. Units and weighting remain protocol-specific.

Track	Released status	Present evidence	Remaining limitation
CIFAR-10-C	replayable diagnostic	per-cell Δ^{cell}, $\widehat{\Delta}$, radius, action; 5 RNG seeds; 3 candidates	opened, dependent, constructed grid from one checkpoint; no untouched family
ImageNet-C	replayable diagnostic	per-condition records for 5 seeds and Tent/EATA/SAR	opened 27-cell controlled configuration; candidate-specific, not one policy
Office-Home	replayable	canonical primary and 54-cell stream-seed records; later five-checkpoint audit reported separately	full-fit-versus-OOF transfer-stability premise not predeclared
iWildCam	invalid archived diagnostic	opened test-split records scored with the wrong macro-F1 definition	no completed current-protocol or official-metric result; population and overlap were not sealed
Camelyon17	replayable diagnostic	archived opened OOD row plus separate B–v2 diagnostics	OOD row is one-sided and non-prospective; B–v2 is not untouched-domain evidence
RxRx1	replayable	displayed model-seed-0 row plus three model-checkpoint records and source hashes	all three checkpoints freeze throughout; no helpful test conditions
CCT-20	release-complete; cell-outcome-unopened, internally sealed execution	5 independent checkpoints, 9 target locations, actions, predictions, score, manifest, and receipts	aggregate target-label metadata had previously been inspected; not globally label-unopened or publicly preregistered; only two fit locations; no ADAPT exposure; trans-target extrapolation remains an assumption
So2Sat-LCZ42	sealed development stop	Tent and SAR candidate bundles; multi-city, multi-checkpoint gate-fit screen; no gate-calibration or target access	no feasible candidate; gate-fit outcomes are open; no gate calibration or target result
PACS	partial	three-seed aggregate and source cross-checks	missing per-cell $\widehat{\Delta}$ and residuals prevent full gate replay
ImageNet-R	replayable	4 seeds, 10 candidate backbones, 12 conditions each	architecture aggregate is diagnostic, not one deployable router
CIFAR-10.1	replayable	locked cross-seed condition records	low-margin conservative match; selective routing not exercised
Physical camera	protocol only	dashboard, logging schema, and planned session split	no completed fresh recording, held-out session, or result table

G.1 CCT-20 Inference, Location, and Release Ledger

Provenance ledger. This was a cell-outcome-unopened, internally sealed execution: individual cell outcomes remained unopened until one-shot scoring. Aggregate target-label metadata had previously been inspected, so the study was not globally label-unopened or publicly preregistered. Item-level target labels were deserialized once, after the one-shot scoring marker was spent. The release chain also records the numeric-v2 repair in a sealed addendum created before outcome access. That repair replaced an

Table 11: Locked CCT-20 primary contrasts over 45 equal-weight checkpoint–location cells from five independent checkpoints and nine location clusters. The estimand is baseline regret minus KGA regret, so positive values favor KGA. A paired two-way product bootstrap resamples checkpoint rows and location columns. Two nominal 97.5% intervals target a nominal 95% Bonferroni family level; actual family coverage still requires valid marginals. Reference p-values enumerate one-sided sign flips of nine location means, with Holm adjustment. This cell-outcome-unopened execution reproduces the locked calculation but does not establish a population guarantee.

Comparator	Baseline regret − KGA regret	Nominal 97.5% CI	Sign-flip p	Holm p	Holm flag (.05)
Always adapt	0.18190227	[0.08501071, 0.31066374]	0.00195312	0.00390625	yes
Always freeze	0	[0, 0]	1	1	no

unstable divergence calculation with a stable Gaussian KL computation; it did not change the model, data split, or scoring rule. We treat it as protocol provenance, not result-driven tuning. All primary outcomes and action/effect counts come from `paper/generated/cct20_release_manifest.json`. Its sidecar `paper/generated/cct20_release_manifest.json.receipt.json` binds the file-level digest; the generated macros instead record canonicalized inference-document SHA-256 `5d1ba74d9a8b16ef24746ca3d95969e635d34fd4112806f6bf91b4b45a6cf907`.

Locked thresholds. Strong success required all six conditions: all 45 cells complete; both nominal 97.5% Bonferroni intervals strictly above zero; both exact nine-location sign-flip tests surviving Holm at 0.05; ADAPT rate at least 0.10; FREEZE rate at least 0.10; and total commitment at least 0.30. Completion, freeze exposure, and commitment passed. The interval and test requirements failed against always-freeze, and ADAPT exposure was zero. The release therefore fails the strong-success criterion.

The nominal pointwise 95% intervals in the CCT utility-preservation table in the main paper belong to the locked utility-preservation endpoint. The nominal 97.5% Bonferroni intervals in the strong-success audit in Table 11 belong to a separate, stricter two-comparison criterion; they are not alternate versions of the same interval.

Finite-sample sign-flip validity requires the null joint law of the nine location gaps to satisfy the main paper's coordinatewise sign-flip invariance condition. Independent zero-symmetric location gaps conditional on the evaluated checkpoints would suffice, but that assumption is unverified here. Exchangeability alone does not suffice. The sign-flip analysis conditions on the evaluated checkpoints, whereas the two-way bootstrap also resamples checkpoint levels. These are different uncertainty models, and neither establishes distribution-free safety for unseen locations.

Primary result. KGA exactly matched always-freeze. Its contrast against always-freeze was 0 with nominal Bonferroni-adjusted 97.5% bootstrap interval [0, 0], enumerated location sign-flip $p = 1$, and Holm $p = 1$. Against always-adapt, the contrast was 0.18190227 with interval [0.08501071, 0.31066374], sign-flip reference $p = 0.00195312$, and Holm $p = 0.00390625$. Measured cell benefit is adapted accuracy minus frozen accuracy: 1 cell was helpful, 0 were exactly zero, and 44 were harmful. These counts explain why freezing was useful here. They do not show that the gate learned when to adapt. KGA made 0/44/1 ADAPT/FREEZE/ABSTAIN decisions. All 45 served predictions used the frozen model, but the sole helpful cell received an interval-supported FREEZE, yielding one false FREEZE among 44 FREEZE actions (1/45 overall). This is all-frozen fail-closed service. It passes only the locked utility-preservation rule and does not demonstrate selective routing or strong success.

Limits and next step. All 55 development cells showed lower performance after Tent than under the frozen model, so the gate had no helpful ADAPT example. The ridge fit used 10 checkpoint–location

Table 12: Complete cell-outcome-unopened CCT-20 location ledger. A/F/U denotes ADAPT/FREEZE/ABSTAIN, and +/0/− counts positive/zero/negative measured-benefit cells. Difference columns are equal-weight means over five checkpoint cells; positive policy differences favor KGA. Values are rounded to four decimals, while the sealed manifest retains full precision. Cells are repeated-model observations, and the inferential cluster unit is the camera location.

Location	n/checkpoint	A/F/U	Benefit +/0/−	Adapt − freeze	KGA − adapt	KGA − freeze
0	1302	0/5/0	0/0/5	-0.5418	0.5418	0.0000
7	1283	0/5/0	1/0/4	-0.0293	0.0293	0.0000
28	911	0/5/0	0/0/5	-0.0562	0.0562	0.0000
40	800	0/5/0	0/0/5	-0.1232	0.1232	0.0000
46	4432	0/4/1	0/0/5	-0.0723	0.0723	0.0000
78	1633	0/5/0	0/0/5	-0.0839	0.0839	0.0000
100	3441	0/5/0	0/0/5	-0.2035	0.2035	0.0000
105	1540	0/5/0	0/0/5	-0.2892	0.2892	0.0000
130	983	0/5/0	0/0/5	-0.2376	0.2376	0.0000

rows from only two camera units for 11 recorded features. Calibration used 45 rows collapsed to nine calibration camera locations, while the locked targets came from held-out target locations. Coverage therefore needs an unverified cross-split exchangeability assumption. Target Mahalanobis diagnostics ranged from 7.12 to 485.67 (median 28.73), showing strong extrapolation; this diagnostic never changed an action and is not a formal coverage guarantee. One natural target study also cannot establish transfer of selective routing across natural shifts. Cell-level 16-indicator macro-F1 is archived as descriptive secondary evidence; no post-hoc aggregate or inference claim is made.

Table 12 reports all nine location clusters. Evaluation n is the number of evaluation images per checkpoint at that location. Each A/F/U triplet and each helpful/zero/harmful triplet therefore has denominator five, one cell for each checkpoint. These labels describe measured cell outcomes. Accuracy differences use one sign throughout: adapted minus frozen for adaptation benefit, and KGA minus the named fixed policy for the two policy columns.

The release ledger contains all 45 cells: 0 ADAPT, 44 FREEZE, and 1 ABSTAIN actions; 1 helpful, 0 zero, and 44 harmful adaptation effects. The release verdict remains SAFE UTILITY ONLY. The location table is descriptive; sign-flip enumeration uses the nine location means under the reference model in the main paper's coordinatewise sign-flip invariance condition, and no post-hoc aggregate is claimed for the secondary macro-F1 outcome.

H Extended KGA Implementation Contract

H.1 State Isolation

The frozen and candidate predictors must not share mutable normalization buffers, optimizer state, or parameter storage during preview. The candidate is created from a checkpoint, adapted on the unlabeled window, and evaluated without overwriting the active state. Only an adapt decision may replace the active checkpoint. Freeze and abstain discard the temporary state. A proposed deployment test should hash the active parameters before and after each non-adapt decision and require equality.

H.2 Evidence Determinism

Evidence extraction must define preprocessing, batch order, numerical precision, class ordering, normalization constants, and missing-value behavior. Paired features require frozen and candidate outputs on the same samples in the same order. Source statistics and ATC thresholds are immutable calibration artifacts. Evidence schema versions should be explicit rather than inferred from vector length, because two schemas can share a dimension while changing semantics.

H.3 Calibration Identity

The calibration identity should hash the source checkpoint, candidate adapter class, adapted parameter names, optimizer and learning rate, update steps, evidence schema, benefit-model hyperparameters, feature scaler, residual construction, α, and inference-unit definition. The radius belongs to that identity. Reusing it after a candidate or evidence change is not a benign software optimization; it changes the random variable whose residual is being covered.

H.4 Runtime Accounting

A deployment report should time capture/preprocessing, frozen inference, candidate adaptation, candidate inference, evidence extraction, benefit prediction, interval decision, copy creation, and rollback separately. Mean and tail latency should be reported at the decision-window level. Peak memory should include both model states and optimizer buffers. The compact manuscript leaves these cells unfilled because the canonical cross-dataset manifest does not provide one comparable runtime measurement. Omitting an unsupported number is preferable to quoting controller-only latency as end-to-end cost.

I Evidence Inventory

I.1 Protocol-Specific Feature and Model Schemas

The controlled-grid implementation records 11 label-free coordinates with effective rank 9. Two coordinates are deterministic differences:

$$\begin{aligned}\texttt{entropy\_drop} &= \texttt{pre\_entropy} - \texttt{post\_entropy},\\ \texttt{pbal\_drop} &= \texttt{pre\_pbal} - \texttt{post\_pbal}.\end{aligned}$$

The panel therefore contains 11 stored values, not 11 independent information sources. Together they describe confidence, class balance, change from the frozen model, and update size.

The exact common coordinate order is

> `pre_entropy`, `pre_conf`, `pre_pbal`, `post_entropy`, `post_conf`, `post_pbal`, `pbal_drop`, `entropy_drop`, `frac_highconf`, `marginal_KL`, and `update_norm`.

The canonical evidence-contract digest for this ordered schema is `76b3951e77c42bbb1359123abd8d937b087c9c42ba0f14e11e61f9aca132f7f7`. The current controlled cross-fit passes these recorded coordinates directly to the GBRT; it applies no separate feature scaler. Nonfinite evidence or outcomes are rejected, and a split that cannot support the exact rank receives an infinite radius and therefore no commitment.

The shared historical builders use natural-log predictive entropy. They normalize class-marginal entropy by $\log K$, stabilize logarithms at 10^{-9}, define the high-confidence fraction with a fixed candidate-confidence threshold of 0.9, and retain marginal KL on its raw scale. The Office-Home builder uses the analogous definitions with 10^{-12} log stabilization. These constants are part of the recorded schema: changing one creates a new evidence variable even when the coordinate name and dimension remain unchanged.

Table 13: Evidence-schema identity crosswalk. Digests bind ordered names and dimensions for the archived contracts; the CCT row instead records its explicit trace schema and sealed gate document. Preprocessing and invalid-input behavior are part of the protocol, not interchangeable defaults.

Authority	Dim.	Schema identity	Model-side preprocessing	Invalid or unavailable state
Common controlled/archive	11	`76b3951e77c42bbb1359123abd8d937b087c9c42ba0f14e11e61f9aca132f7f7`	raw coordinates to candidate-specific GBRT	controlled cross-fit fail-closes malformed or nonfinite matrices
Camelyon17 OOD	17	`bf6eb06e5048705c02293c85038c732e0aacc6e6737f6cb481b3bc5df83ca746`	raw coordinates to archived GBRT	record validator rejects wrong width or nonfinite values
Office-Home	18	`a363c05a2475f3170976b8f7efc30763859e29b9cfe1714f9c4f91a4f1e892bc`	raw coordinates to archived GBRT	absent source logits use recorded zero fallbacks for energy shift and ATC
RxRx1	11	names unrecovered; `8444fd58264350b2379bff08f5ccc0259f381265db7a03f9e0b48415e562805e`	archived model only; coordinate names are not inferred	validator rejects wrong width or nonfinite values
CCT-20	11	`kbound_cct20_label_free_trace_features_v1`; sealed gate document `74db83cca3641c505735bcc9db4f76c588148f1ebb4d7d7c2b713bfbb6eda0e7`	fit-location mean and population-SD standardization before ridge	exact-schema, finiteness, or support failure returns ABSTAIN

Natural-shift artifacts do not all use the same schema. The Camelyon17 OOD record uses the common 11 coordinates plus `disagreement_rate`, `entropy_gap`, `energy_shift`, `bn_kl`, `atc_acc_est`, and `conf_drop`; its 17-coordinate digest is `bf6eb06e5048705c02293c85038c732e0aacc6e6737f6cb481b3bc5df83ca746`. The separate Camelyon17 B–v2 diagnostic uses the common 11-coordinate schema. Office-Home adds `disagreement`, `energy_shift`, `bn_kl`, `atc_acc_est`, `conf_drop`, `prior_kl_source`, and `cand_disagree_mean`; its 18-coordinate digest is `a363c05a2475f3170976b8f7efc30763859e29b9cfe1714f9c4f91a4f1e892bc`. The RxRx1 authority records dimension 11 and digest `8444fd58264350b2379bff08f5ccc0259f381265db7a03f9e0b48415e562805e`, but explicitly marks the coordinate names as unrecovered. We therefore do not silently assign it the common names. The iWildCam record is withheld from release-level numerical claims because its archived scorer used scikit-learn macro-F1 rather than the official WILDS label-present macro-F1 contract. Its canonical evidence metadata nevertheless recovers the common ordered 11-coordinate schema and digest above.

A fitted model and its radius remain specific to their recorded evidence schema. Missing-input behavior is protocol-specific. The current controlled cross-fit fail-closes malformed or nonfinite numeric matrices to ABSTAIN, and CCT-20 returns ABSTAIN on an exact-schema or finiteness failure. The maintained natural-shift record validator instead rejects wrong-width or nonfinite records before routing; historical feature builders encode unavailable BatchNorm statistics as zero, and the Office-Home builder also defines zero fallbacks for absent source logits in its energy-shift and ATC coordinates.

The controlled and archived GBRT protocols use scikit-learn's `GradientBoostingRegressor` (Pedregosa et al., 2011) with squared-error loss, 250 trees, depth 2, learning rate 0.05, subsample 0.8, and random seed 0. The fit is specific to each dataset and adapter. CCT-20 instead uses its sealed ridge model, as described in the main paper's CCT-20 design. Thus "adapter agnostic" describes the interface: any adapter can propose a candidate that is evaluated and either committed or rolled back. It does not mean that a benefit model fitted for Tent transfers validly to EATA, SAR, or another adapter.

CCT-20 trace and ridge contract. CCT-20 uses a separate ordered trace schema:

> `frozen_mean_entropy`, `tent_mean_entropy`, `entropy_change`, `frozen_mean_confidence`, `tent_mean_confidence`, `confidence_change`, `prediction_disagreement`, `marginal_jensen_shannon_divergence`, `normalized_predicted_class_effective_count`, `normalized_tent_update_norm`, and `batchnorm_batch_source_statistic_divergence`.

The sealed gate identifies this contract as `kbound_cct20_ridge_gate_v1`; the release manifest binds its canonical document by SHA-256 `74db83cca3641c505735bcc9db4f76c588148f1ebb4d7d7c2b713b fbb6eda0e7`. Every feature must be present exactly once and finite. Fit-location means and population standard deviations standardize the coordinates; a zero standard deviation is replaced by one. The ridge penalty is 10, the intercept is unpenalized, and calibration at $\alpha = 0.10$ takes the maximum absolute residual across checkpoints within each location before applying the exact-rank rule over nine locations. A support or schema failure returns ABSTAIN; Mahalanobis distance is diagnostic and never changes the action.

Before ridge standardization, the CCT trace extractor normalizes each entropy by $\log 16$, the Jensen–Shannon divergence by $\log 2$, and predicted-class effective count by 16. It defines the update coordinate as

$$\frac{\|\theta_{\text{after}} - \theta_{\text{before}}\|_2}{\max\{\|\theta_{\text{before}}\|_2, 10^{-12}\}}$$

over the official Tent BatchNorm-affine parameters. The final coordinate is a channel-weighted Gaussian KL divergence from the probe-batch statistics to the stored source BatchNorm statistics, using the layer's BatchNorm epsilon. Probability values are clipped only at machine tiny before a logarithm. These normalizations are distinct from the archived GBRT coordinates above.

Table 14: Non-empirical feature-family summary for the maintained implementation. Each row names one feature family, the state required to compute it, and its protocol-specific validation or fallback. All quantities aggregate unlabeled inputs, frozen/candidate outputs, or the proposed update; the table reports no performance result.

Family	Representative coordinates	Required state	Failure handling
Predictive uncertainty	mean entropy, confidence, quantiles	frozen/candidate softmax	reject NaN or malformed probabilities
Marginal balance	normalized class-marginal entropy and change	predicted-class marginals	fixed class count and normalization
Model change	entropy gap, confidence drop, prediction disagreement	paired frozen/candidate outputs	require aligned sample order
Collapse signal	fraction above adapted confidence 0.9	candidate probabilities	threshold fixed before scoring
Distribution shift	free-energy shift, batch/running-statistic divergence	frozen logits and source statistics	protocol-specific validation or documented zero fallback
Update magnitude	parameter-update ℓ_2 norm	shadow candidate delta	configuration hash must match calibration
Source-calibrated proxy	ATC-style threshold estimate	source-fixed threshold	target labels never used

These protocol-specific dimensions are part of the fitted estimator contract, not universal sufficient statistics. Feature selection may improve prediction, but it cannot defeat the label-kernel construction in the main paper's label-kernel-freedom lemma over an unrestricted target fibre.

J Implementation Failure Modes and Conservative Responses

The implementation must distinguish a supported negative interval from an unavailable assessment. Detectable identity, schema, support, or calibration failures therefore retain f_0 and record ABSTAIN unless a predeclared recovery rule restores a validated state.

Table 15: Non-empirical deployment checklist. Each row maps one detectable operational failure to the conservative response required by the implementation contract; it reports no benchmark outcome.

Failure	Detection	Response
Checkpoint/adapter mismatch	stored hash mismatch	rollback and recalibrate
Evidence outside support	distance or support audit	abstain
Missing/invalid feature	schema and finite-value check	retain f_0; abstain
Insufficient calibration size	rank $k > n$	infinite radius
Batch too small	minimum-size contract	wait or abstain
Residual transfer concern	group/residual diagnostic	widen or withhold claim
Continual-state drift	state and stream monitor	restore the last validated state

The audit log should store a timestamp, dataset or stream identifier, checkpoint and adapter hashes, evidence schema version, evidence vector, benefit estimate, radius, interval endpoints, action, rollback outcome, latency components, and configuration hash. Labels, if later revealed, are joined in a separate offline audit table. This separation makes it possible to verify that the live decision was target-label-free.

K Prospective Natural-Shift Confirmation Protocol

CCT-20 completed a cell-outcome-unopened, internally sealed natural-shift evaluation but produced all-frozen fail-closed service, not selective routing (Section G.1). So2Sat-LCZ42 then stopped at its development-feasibility gate before gate calibration or target access (Section B.10). Its city-level outcome summaries are now open, and its declared city-level target action conflicts with the runner's checkpoint–city action unit. A new confirmation must therefore be versioned independently, exclude those opened outcomes from tuning, and resolve the action unit before target execution. The design below is future evidence, not a reinterpretation of either completed record.

K.1 Lock Before Target Labels

The protocol should name one source checkpoint, one candidate adapter, one evidence schema, one benefit-model class, one calibration rule, one α, and one target-domain family. Adapter hyperparameters must be selected on development domains and then frozen. The lock should hash the checkpoint, code revision, configuration, calibration records, and planned scoring script. It must also name the test unit and the multiplicity family. A later change creates a new protocol version; it does not overwrite an earlier development-stopped or incomplete arm.

Natural heterogeneity should be built through independently defined environments rather than by pooling whichever benchmark conditions produce opposite signs. Examples include hospitals, camera locations, laboratory batches, acquisition days, physical devices, or visual domains. Environment membership may be observed at deployment, but the measured cell-benefit label remains hidden. The target test environments should be absent from estimator fitting, residual calibration, threshold tuning, and candidate selection.

K.2 Three-Way Data Separation

Development data fit h_θ and select the evidence pipeline. A separate calibration set produces residuals for the main paper's exact-rank radius rule. The held-out target set is opened exactly once after all software and configuration hashes match the lock. Splitting adjacent frames, patches, or samples from the same session across these stages is leakage even when filenames differ. The split must occur at the environment or session level that defines deployment exchangeability.

The target stream is replayed identically for always-freeze, always-adapt, simple gates, KGA without a radius, and the full empirical benefit-interval gate. Model randomness should either be frozen for paired replay or repeated through independently trained checkpoints declared in advance. Every policy receives the same ordered windows. Labels are stored behind a separate audit interface and are unavailable to the live decision process.

K.3 Endpoints and Decision Rule

The primary endpoint is paired regret against each fixed policy. Success requires both baseline minus KGA regret gaps to be positive and both predeclared intervals to exclude zero, with coverage justified for the declared estimand and sampling model. If the intended claim concerns population risk rather than measured cells, the protocol must also justify that transfer. The false-adapt endpoint is $\mathrm{FA_u}$ with the adapt count reported beside it. Secondary endpoints are $\mathrm{FA_c}$, adapt rate, freeze rate, abstain rate, commitment rate, task accuracy, and latency. Conditional false-adapt is descriptive unless a separate selective-risk result is proved.

The primary benefit-interval rule remains unchanged: adapt if $\widehat{\Delta} - \varepsilon > 0$, freeze if $\widehat{\Delta} + \varepsilon < 0$, and abstain otherwise. The experiment must not lower ε, change α, or select a different benefit regressor after observing target labels. Exploratory sensitivity analyses may be reported, but they remain outside the primary bar.

Table 16: Protocol requirements for a future confirmatory natural-shift KGA experiment. Each row states what must be frozen before target labels are revealed and the corresponding audit record; the table is a design specification, not a completed-study result.

Component	Must be fixed	Audit evidence
Model and candidate	checkpoint, adapter, optimizer, steps, adapted parameters	hashes and immutable configuration
Evidence and estimator	schema, normalization, model class, hyperparameters, seed	feature contract and fitted-artifact hash
Calibration	split unit, residual construction, exact rank, α	calibration IDs, residuals, radius, no target overlap
Target evaluation	environments, windowing, order, exclusion rules, transductive/inductive candidate semantics	sealed manifest and one-time scoring log
Primary inference	two paired regret gaps, cluster unit, interval method	code and random seed fixed before label reveal
Safety	$\mathrm{FA_u}$ and adapt exposure	action log joined to labels only offline
Multiplicity	all candidate–track arms in family	time-stamped inventory including failures
Runtime	end-to-end components and peak memory	raw timing rows on the target hardware

K.4 Interpreting Outcomes

Four outcomes are scientifically useful. An interval-supported reduction in regret relative to each fixed comparator, with nontrivial ADAPT and FREEZE exposure, supports routing utility under the locked natural-shift class. Matching the better fixed policy with few errors supports conservative commitment but not selective routing. High abstention with low error indicates a conservative interval gate and motivates better evidence or more calibration data. A comparator shortfall fails the predeclared routing-utility criterion; excess false adaptation challenges the corresponding safety or calibration criterion. Both outcomes must remain in the result family. None of these outcomes alone establishes the population band: that conclusion still requires the declared class and its structural assumptions.

L Theorem Dependencies, Counterexamples, and Nonclaims

The compact paper uses three related logical statements. First, the disagreement reduction is an identity under the binary setup. Second, the population frontier is a uniform statement over a rich, declared class. Third, the empirical coverage-to-action implication is marginal and conditional on interval coverage. The population statements concern Δ; the reported empirical calibration concerns Δ^{cell}. Connecting these targets requires an additional validity argument. Neither the assumptions nor the quantifiers are interchangeable.

Why the multiclass interface does not extend the binary theorem. Under multiclass 0/1 loss, $\Delta = \mathbb{P}_T(D)(p_a - p_0)$, where $p_a = \mathbb{P}_T(f_a = Y \mid D)$ and $p_0 = \mathbb{P}_T(f_0 = Y \mid D)$. Only the binary setting forces $p_0 = 1 - p_a$. KGA may predict a multiclass scalar score difference, but that interface does not inherit the binary population identity.

Table 17: Logical dependency map for the paper's formal and empirical statements. Each row lists a statement's inputs, conclusion, and excluded inference; it reports no additional empirical result, and a check in one row does not validate premises in another row.

Statement	Inputs	Conclusion	Not concluded
Disagreement reduction	binary labels, 0/1 loss, $\mu_T(D) > 0$	$\Delta = 2\mu_T(D)(M + \gamma)$	calibration transfer or observability of γ
Interior impossibility	label-free fibre, $\beta > 0$, $\lvert M\rvert < \beta$, class richness	evidence-identical laws with opposite nonzero benefits	every empirical abstention is structural
Closed-band result	strict directional semantics and boundary law	no uniformly sound strict action on $\lvert M\rvert \le \beta$	adapt/freeze have different task risk at $\Delta = 0$
Frontier sufficiency	declared $\lvert\gamma\rvert \le \beta$ and fixed M	direction is fixed when $\lvert M\rvert > \beta$	label-free estimation of β
Coverage-to-action proposition	marginal interval coverage for a declared target B	marginal false-adapt/freeze control for B	conditional error control or risk alignment
KGA benchmark	fitted h_θ, declared residual rule, frozen protocol	measured regret and actions on saved units	population-benefit coverage or universal transfer to a new domain
Lean build	encoded definitions and assumptions	kernel-checked formal statements	correctness of unencoded data and deployment assumptions

Why the boundary is separate. For $|M| < \beta$, the construction chooses two latent residuals that yield opposite nonzero benefits. At $|M| = \beta > 0$, one admissible residual can make the benefit zero while another retains the sign of M; opposite nonzero signs need not exist. Both cases block a uniform strict commitment, but for different reasons. At $M = \beta = 0$, every law in the zero-residual class has zero benefit under the model. Abstention remains the maximal action only under the paper's strict-direction convention, not because either task-risk choice is worse.

Why a calibration residual need not be drift. Let $P_T = P_S$, with equiprobable inputs a, b, deterministic labels $Y(a) = 1$, $Y(b) = 0$, and candidate $f_a \equiv 1$. Set $f_0(a) = 0$, $f_0(b) = 1$, and $s \equiv \frac{1}{2}$. The candidate is correct on half the source inputs, so this constant score is source-calibrated in the ordinary sense. Yet disagreement occurs only at a, where the candidate is always correct. Thus $M = 0$, $\gamma = \frac{1}{2}$, and $\Delta = \frac{1}{2}$, although the source and target laws are identical. Conditioning on disagreement, not a distribution change, explains the nonzero residual. The frontier requires a bound on this residual regardless of its cause. Top-label calibration conditions on the predicted class (Gupta and Ramdas, 2022); neither it nor ordinary score calibration automatically implies calibration conditional on disagreement.

Why coverage is the premise. The coverage-to-action proof is elementary once the main paper's measured-cell coverage premise holds: on the coverage event, an interval wholly above zero cannot contain a nonpositive Δ^{cell}. The scientific burden is therefore to justify coverage for the declared outcome and calibration/deployment pair. Joint exchangeability of the residual scores is one route; known shift correction or another valid construction may be another. Risk alignment can explain why a useful predictor transfers, but it is not an additional logical premise after coverage is given.

Why cell coverage is not population coverage. Consider the fixed binary predictors $f_0 \equiv 0$ and $f_a \equiv 1$. Let the target input be a with probability 0.45 and b with probability 0.55, with deterministic labels $Y(a) = 1$ and $Y(b) = 0$. Population benefit is $\Delta = -0.10$. In a one-item evaluation cell, a fixed predictor that maps a to $+1$ and b to -1 predicts Δ^{cell} perfectly without reading the new label. With radius zero, cell coverage is perfect, yet the gate selects ADAPT with probability 0.45 despite negative benefit under the fixed population law. This constructed example does not contradict the main paper's coverage-to-action proposition; it shows why its covered target cannot be changed.

Why exchangeability alone does not validate sign flips. As a separate null example, let all nine cluster gaps equal the same fair random sign S. This vector is exchangeable, each gap is symmetric about zero, and a global sign change preserves its law. Nevertheless, when $S = +1$, independent-coordinate sign-flip enumeration for the mean gives a reference value of 1/512. That occurs with probability 1/2, not 1/512. The vector fails the main paper's coordinatewise sign-flip invariance condition. This example explains the required assumption; it is not a model fitted to the CCT-20 observations.

Why a null is not an impossibility result. A null benchmark row can arise because the true effects are small, the candidate is one-sided, the evidence map is weak, the benefit model underfits, the radius is wide, or calibration does not transfer. The population impossibility requires evidence-identical target laws inside the declared class with incompatible strict directions. The canonical benchmark table does not estimate that full fibre. It therefore reports weak or adverse empirical regimes without calling them proven unknowable worlds.

What would falsify the practical claim. The practical claim is conditional and testable. A clean target replay with the locked pipeline can show excessive FA_{u}, poor coverage, or regret above both fixed policies. A checkpoint or adapter change can show that calibration does not transfer. A family-held-out split can show that cell-level cross-fitting was optimistic. Such failures do not refute the population identity; they refute the operational assumptions used to obtain a useful empirical interval.

M Formalization and Artifact Scope

The Lean project under `docs/research/kbound/formal/` provides a kernel-checked inventory for the theorem names it exports. The compact paper relies on the LaTeX statements above; it does not claim that every sentence has a one-to-one Lean theorem.

In particular, deployment-class richness, risk alignment of a learned evidence map, exchangeability of a benchmark split, and correctness of data preprocessing remain outside the proof assistant unless separately encoded.

The registered audit contains 142 declarations: the 65-result finite and algebraic core plus 77 probability capstones and counterexample results. The added modules derive one-shot residual coverage from exchangeable measurable scores, including ties; filtered Ville and bounded optional-stopping bounds; general KL/TV testing bounds for finite product experiments; and independent bounded and martingale-difference concentration inequalities. Their assumptions are explicit. In particular, these results do not establish exchangeability or conditional nulls for the benchmark data, nor do they turn batch-level coverage into population-risk coverage.

The measurable target-law construction also goes beyond the original two-atom example. It fixes the input law and off-disagreement label kernel, varies a measurable correctness field on disagreement, and integrates the actual zero-one loss difference. With labels supported on the two predictions on disagreement, feasible margins, and positive disagreement mass, it proves the clipped identified interval and exact strict ADAPT/FREEZE frontiers without assuming `RichAt`. It does not prove richness for an arbitrary restricted deployment subclass.

One historical extension remains outside the verified scope. A measurable label swap preserves label-free evidence and reverses population benefit, but selecting one representative from each swap orbit need not identify the sign: distinct orbits can share the same evidence law and have opposite selected signs. Lean verifies this counterexample and the corresponding set-theoretic fibre-consistency criterion, not a general measurable decoder or the older H/ratio-rate extension. Those stronger claims are excluded from this compact submission. The `-full-foundations` gate therefore remains failing for the sixth layer; the scoped kernel audit checks the registered results and rejects nonstandard proof axioms.

The formal build and empirical build answer different questions. Lean checks that a proposition follows from formal assumptions. The reconciliation scripts check that displayed aggregates match saved records under implemented definitions. Neither checks the truth of an unencoded deployment assumption, and neither substitutes for independent experimental replication.

The authoritative compact artifacts are therefore layered:

1. this source and its compiled PDF for the scoped narrative;
2. shared theorem inputs for the exact statements and proofs;
3. canonical JSON and generated LaTeX for numerical results;
4. the source manifest for record-level provenance;
5. package tests and Lean logs for implementation and formal checks.

Historical drafts, dashboards, and exploratory search folders can remain useful development records, but they are not evidence sources for this compact submission unless referenced by the canonical manifest.

N Reproducibility and Release Procedures

N.1 Canonical Numerical Source of Truth

The CIFAR rows and appendix reconciled point estimates share one numerical source: `experiments/kbound/results/reconciled_panels_v1/canonical_panel_results.json`, whose SHA-256 is `35d4c165843de1ece3cb35ffb4ac50dbfcbfa33b646754e2b6d133fbdfa78c6e`. Its input inventory is bound separately by source-manifest SHA-256 `03b1d2b1e9e5ed1cf835126f871ed83eb8497a1ac81727b3955af0d69eb89742`; the canonical JSON also records generator, implementation-dependency, and per-array bindings. Auxiliary artifacts retain narrower roles. The family analysis is the retrospective sensitivity in Table 7; the POEM/AETTA artifact is historical; and the So2Sat record is a locked development stop. The generated result manifest indexes each reported claim and scope. None of these auxiliary records overwrites the canonical point-estimate panel.

N.2 Reproduction Pipeline

The full gate uses Python 3.12 and `requirements-research.txt`. The smaller paper-only profile does not include the Torch imports used during repository-wide test collection. The model runtime uses PyTorch (Paszke et al., 2019). From the repository root, the release command is:

```
KBOUND_PYTHON=.venv/bin/python bash
docs/research/kbound/runbooks/release_candidate.sh all
```

It runs the read-only preflight, validates and regenerates result authorities, executes software and formal checks, builds the four current role PDFs, renders every PDF page, and emits release checksums. A DOCX may be produced as a nonrelease convenience artifact, but it is not a fifth current role. The gate consumes frozen model outputs and never launches training.

For presentation-only rebuilding from already validated authorities, the command is:

```
SOURCE_SNAPSHOT_COMMIT=<12-hex-source-commit>
BUILD_LONG_TMLR=1 BUILD_SHORT_MAIN=1
BUILD_SHORT_SUPPLEMENT=1 BUILD_FULL_REPORT=1 bash
docs/research/kbound/scripts/build_pdfs.sh
```

This shorter command validates the frozen authorities, regenerates presentation tables and figures, and builds the four maintained role PDFs. It does not replace the full release gate or create new empirical evidence.

N.3 Calibration Identity and Implementation Contract

The maintained API binds an interval-gate record to the checkpoint, adapter configuration, evidence schema, benefit estimator, residual construction, safety level, and scoring unit. The deployment path must fail closed when any identity differs. `KGA.certify_evidence` requires a frozen estimator, the exact schema, a protocol hash, and a separate residual set. The lower-level `certify` function is an audit interface; it does not construct evidence or validate transfer by itself.

N.4 Formalization and Lean Scope

Lean artifacts and validators check the encoded theorem cores and selected finite constructions. Formalization of learning-theory results has precedent, including a Rademacher-complexity generalization bound (Sonoda et al., 2025); we cite its pinned 2025 version as context, not as a proof of K-Bound's assumptions. These checks do not certify dataset correctness, preprocessing, empirical transfer, exchangeability, or every deployment assumption stated in prose. Machine checking strengthens the statements it encodes; it does not validate the benchmark pipeline outside those statements.

N.5 Release Checklist and Audit Logs

The Phase-1 provenance audit matches all 106 canonical source/compact hash pairs, all 10 recoverable configuration hashes, and all 6 archived checkpoint hashes; it also records a 21-file current-code snapshot. Historical runs that did not serialize full identity remain explicitly partial rather than being backfilled. The release gate runs reconciliation tests, claim-ledger and result-manifest validation, manuscript checks, LaTeX builds, checksum verification, and visual PDF inspection. Build logs, the canonical manifest, the provenance seal, and the final PDFs are archived together.

Table 18: Release-stage audit contract. Each stage has an explicit authority and a fail-stop condition, preventing a mixed-generation or partially verified manuscript set from being published as current.

Stage	Checked authority	Failure response
Numerical reconciliation	canonical panel, source manifest, CCT manifest and receipt	stop on hash or claim mismatch
Source binding	reviewed source snapshot and release-role map	stop on missing or ambiguous identity
Compilation	four maintained TeX drivers from one source closure	stop on compile, reference, or overflow defect
Structure and anonymity	page structure, fonts, metadata, links, and anonymous TMLR author fields	reject malformed output or identity leakage
Visual inspection	fresh rendering of every maintained PDF page	repair clipping, blank regions, orphaned captions, or displaced floats
Publication	exact four-PDF inventory and SHA-256 manifest	publish atomically; never mix generations

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